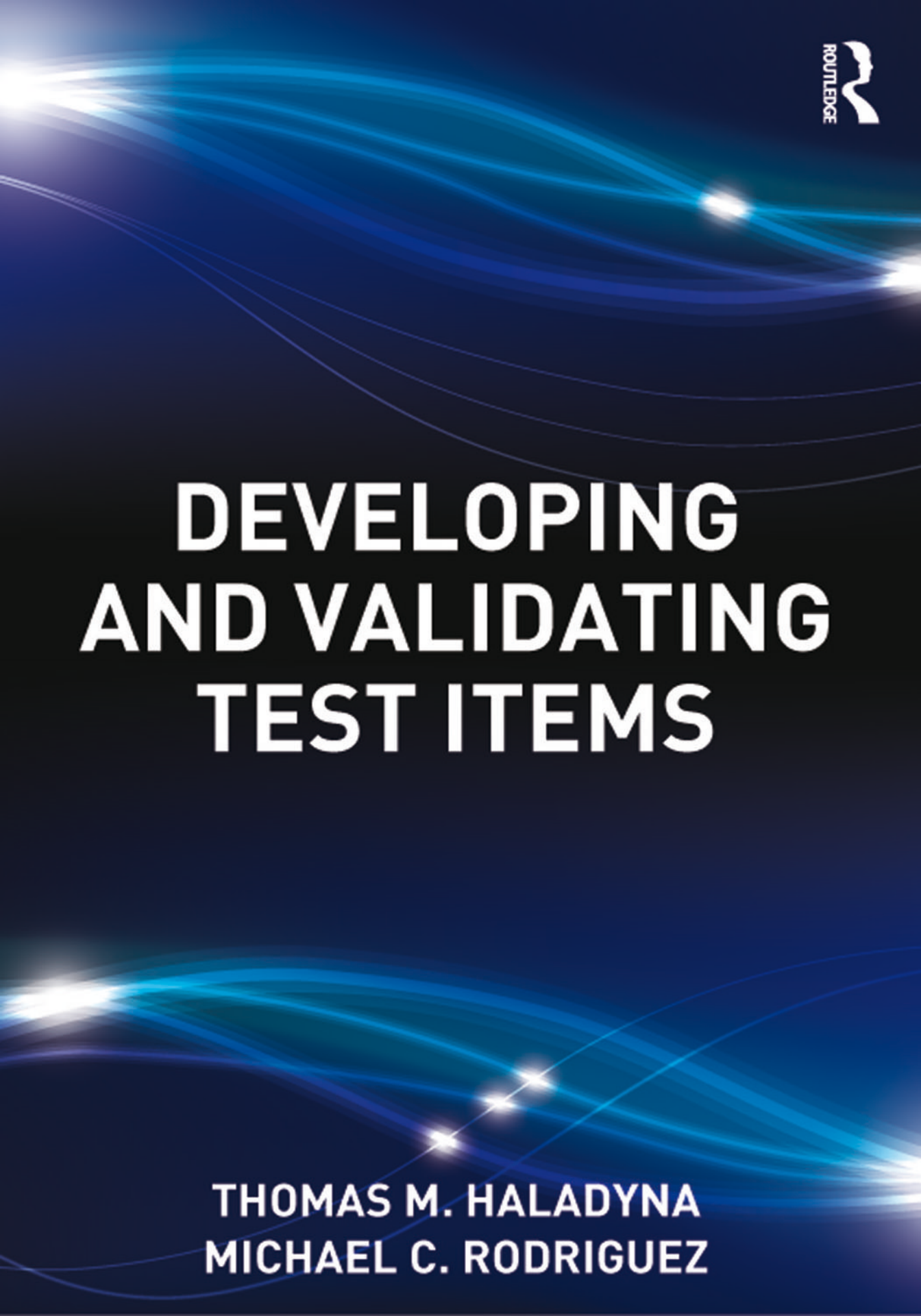

ROUTLEDGE
DEVELOPING AND VALIDATING TEST ITEMS
THOMAS M. HALADYNA
MICHAEL C. RODRIGUEZ

DEVELOPING AND VALIDATING TEST ITEMS

Since test items are the building blocks of any test, learning how to develop and validate test items has always been critical to the teaching–learning process. As they grow in importance and use, testing programs increasingly supplement the use of selected-response (multiple-choice) items with constructed-response formats. This trend is expected to continue. As a result, a new item-writing book is needed, one that provides comprehensive coverage of both types of items and of the validity theory underlying them.

This book is an outgrowth of the co-author's previous book, *Developing and Validating Multiple-Choice Test Items*, 3rd Edition (Haladyna, 2004). That book achieved distinction as the leading source of guidance on creating and validating selected-response test items. As with its predecessor, the content of this new book is based on both an extensive review of the literature and on its author's long experience in the testing field. It is very timely in this era of burgeoning testing programs, especially when these items are delivered in a computer-based environment. Key features include:

Comprehensive and Flexible—No other book so thoroughly covers the field of test item development and its various applications.

Focus on Validity—Validity, the most important consideration in testing, is stressed throughout and is based on the Standards for Educational and Psychological Testing, currently under revision by AERA, APA, and NCME

Illustrative Examples—The book presents various selected- and constructed-response formats and uses many examples to illustrate correct and incorrect ways of writing items. Strategies for training item writers and developing large numbers of items using algorithms and other item-generating methods are also presented.

Based on Theory and Research—A comprehensive review and synthesis of existing research runs throughout the book and complements the expertise of its authors.

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DEVELOPING AND VALIDATING TEST ITEMS

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Preface

Although the scholarly study of item development has been ongoing for quite some time, critics have often noted that this study does not match the effort we give to statistical theories and methods in testing. This book documents the progress we have made in the science of item development but, at the same time, issues warnings and offers suggestions about future efforts.

Our goal is to provide readers with a comprehensive, authoritative volume on how to develop all kinds of test items and how to ensure that when an item is used in a test it performs as it should. We refer to the process of getting an item ready for a test as *item validation*.

This book has been a long collaborative labor that has been evolving over many years.

Earlier versions of this book were mainly aimed at the selected-response formats. This book examines constructed-response formats and also includes some special topics that seem justified by their popularity and importance in testing. We have attempted to provide up-to-date information from the most authoritative sources and also drawing from our collective experiences.

The book is organized into six parts. The first part presents foundation information about validity, the item development process, the challenging problem of defining item content and cognitive demand, and the equally difficult choice of item formats. The second part deals exclusively with the selected-response format, with the exception of chapter 9. This chapter was added to provide guidance on the development of survey items. The third part is complementary to the second part. This part offers a variety of constructed-response formats, guidelines, and information on scoring. Part IV deals with three unique areas. Each chapter addresses the problems, research, and approaches to measurement in one of these areas. The areas are writing, credentialing and exceptionalities. We could have expanded this part of the book, as we think these three are very important and challenging for item and test developers. Each chapter provides useful information and recommendations for both research and more effective item development.

Part V deals with item validation. The four chapters are intended to be complementary, but some overlap is intentional because of the importance of procedures and statistical study of item development and item responses. Part VI has a single chapter—a prospective appraisal of where we are and where we need to go to advance the science of item development and validation.

We hope this volume meets your needs, and future editions will only expand on these efforts.

T. Haladyna
M. Rodriguez

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I
A Foundation for Developing and Validating Test Items

Part I covers four important, interrelated concerns in item development and validation.

This first chapter provides definitions of basic terms and distinctions useful in identifying what is going to be measured. The first chapter also discusses validity and the validation process as it applies to item development. The second chapter presents the essential steps in item development and validation. The third chapter presents information on the role of content and cognitive demand in item development and validation. The fourth chapter presents a taxonomy of selected-response (SR) and constructed (CR) test item formats for certain types of content and cognitive demands.

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1
The Role of Validity in Item Development

Overview

This chapter provides a conceptual basis for understanding the important role of validity in item development. First, basic terms are defined. Then the content of tests is differentiated. An argument-based approach to validity is presented that is consistent with current validity theory. The item development process and item validation are two related steps that are integral to item validity. The concept of item validity is applied throughout all chapters of this book.

Defining the Test Item

A test item is a device for obtaining information about a test taker's domain of knowledge and skills or a domain of tasks that define a construct. Familiar constructs in education are reading, writing, speaking, and listening. Constructs also apply to professions: medicine, teaching, accountancy, nursing, and the like. Every test item has the same three components:

1. Instructions to the test taker,
2. Conditions for performance, and
3. A scoring rule.

A *test item* is the basic unit of observation in any test. The most fundamental distinction for the test item is whether the test taker chooses an answer (selected-response: SR) or creates an answer (constructed-response: CR). The SR format is often known as *multiple-choice*. The CR format also has many other names including *open-ended, performance, authentic*, and *completion*. This SR–CR distinction is the basis for the organization of chapters in this book. The response to any SR or CR item is scorable. Some items can be scored dichotomously, one for right and zero for wrong, or polytomously using a rating scale or some graded series of responses. Refined distinctions in item formats are presented in greater detail in chapter 4.

Thorndike (1967) advised item and test developers that the more effort we put into building better test items, the better the test is likely to be. To phrase it as to validity, the greater effort expended to improve the quality of test items in the item bank, the greater degree of validity we are likely to attain. As item development is a major step in test development, validity can be greatly affected by a sound, comprehensive effort to develop and validate test items.

Toward that end, we should develop each test item to represent a single type of content and a single type of cognitive behavior as accurately as is humanly possible. For a test item to measure

multiple content and cognitive behaviors goes well beyond our ability to understand the meaning of a test taker's response to such an item.

Defining the Test

A *test* is a measuring device intended numerically to describe the degree or amount of a construct under uniform, standardized conditions. Standardization is a very important idea when considering a test and the most important feature of a test is the validity of its test score interpretation and use. "Measurement procedures tend to control irrelevant sources of variability by standardizing the tasks to be performed, the conditions under which they are performed, and the criteria used to interpret the results" (Kane, 2006b, p. 17).

In educational achievement testing, most tests contain a single item or set of test items intended to measure a domain of knowledge or skills or a domain of tasks representing an ability. The single test item might be a writing prompt or a complex mathematics problem. Responses to a single test item or a collection of test items are scorable using complex scoring guides and highly trained raters. The use of scoring rules helps to create a *test score* that is based on the test taker's responses to these test items. In this book, we are less concerned with tests and solely concerned with developing highly effective items and then assembling validity evidence for each item response's valid interpretation and use. Readers are directed to the *Handbook of Test Development* (Downing & Haladyna, 2006) for comprehensive discussions of issues and steps in the test development process. The fourth edition of *Educational Measurement* (Brennan, 2006) also provides current treatments of many important issues in test development and validation.

What Do Tests and Test Items Measure?

In this section, two issues we face in the measurement of any cognitive ability are presented and discussed. The first is the dilemma provided when we fail to define a construct operationally that we want to measure. The second is a distinction between achievement and intelligence.

A *construct* is something definable that we want to measure. Constructs have characteristics that help define it. Another good way to make a construct clear is to list examples and nonexamples. In educational and psychological testing, the most important concepts we measure include reading, writing, speaking, listening, mathematical problem-solving, scientific problem-solving, and critical thinking as applied in literature analysis and in social studies. Some concepts are subject-matter-based, for example language arts, mathematics, science, social studies, physical education, and English language proficiency. Professional competence is another type of concept that we often test for certification and licensure. Medicine, nursing, dentistry, accountancy, architecture, pharmacy, and teaching are all constructs of differing professional competence.

Operational Definitions and Constructs

Operational definitions are commonly agreed on by those responsible and most highly qualified for measuring the construct. In other words we have a consensus by highly qualified subject-matter experts (SMEs). In the *Conduct of Inquiry*, it was stated:

> To each construct there corresponds a set of operations involved in its scientific use. To know these operations is to understand the construct as fully as science requires; without knowing them, we do not know what the scientific meaning of the construct is, not even whether it has scientific meaning. (Kaplan, 1963, p. 40)

With an operational definition, we have no surplus meaning or confusion about the construct. We can be very precise in the measurement of an operationally defined construct. We can eliminate or reduce random or systematic error when measuring any operationally defined construct. Instances of operationally defined constructs include time, volume, distance, height, speed, and weight. Each can be measured with great precision because the definition of each of these constructs is specific enough. Test development for any construct that is operationally defined is usually very easy.

However, many constructs in education and psychology are not amenable to operational definition. Validity theorists advise that the alternative strategy is one of defining and validating constructs. By doing so, we recognize that the construct is too complex to define operationally (Cronbach & Meehl, 1955; Kane 2006b; Kaplan, 1963; Messick, 1989). As previously noted, constructs include reading and writing. Also, each profession or specialty in life is a construct. For example baseball ability, financial analysis, quilt-making, and dentistry are examples of constructs that have usefulness in society. Each construct is very complex. Each construct requires the use of knowledge and skills in complex ways. Often we can conceive of each construct as to a domain of tasks performed.

For every construct, we can identify some aspects that can be operationally defined. For instance, in writing, we have spelling, punctuation, and grammatical usage that is operationally defined and easily measured. In mathematics, computation can be operationally defined. In most professions, we can identify sets of tasks that are either performed or not performed. Each of these tasks is operationally defined. However, these examples of operational definition within a construct represent the minority of tasks that comprise the construct. We are still limited to construct measurement and the problems it brings due to the construct's complexity and the need for expert judgment to evaluate performance.

Because constructs are complex and abstractly defined, we employ a strategy known as *construct validation*. This investigative process is discussed later in this chapter and used throughout this book. The investigation involves many important steps, and it leads to a conclusion about validity.

Achievement and Intelligence

The context for this book is the measuring of achievement that is the goal of instruction or training. Most testing programs are designed for elementary, secondary, college and graduate school education. Another large area of testing involves certifying professions, such as medicine, dentistry, accountancy, and the like.

Achievement is usually thought of as planned changes in cognitive behavior of students that result from instruction or training, although certainly achievement is possible due to factors outside instruction or training. All achievement can be defined in terms of content. This content can be represented in two ways. The first is a domain of knowledge and skills. The second is as a cognitive ability for which there is a domain of tasks to be performed. Chapter 3 refines the distinctions between these two types of content. However, introducing these distinctions in the realm of achievement is important as we consider item development and validation because it involves validity.

Knowledge is a fundamental type of learning that include facts, concepts, principles, and procedures that can be memorized or understood. Most student learning consists of knowledge. Knowledge is often organized as a domain that consists of an organized set of instructional objectives/content standards.

A *skill* is a learned, observable, performed act. A skill is easily recognized in nearly all settings. For instance, writing skills include spelling, punctuation, and grammar. Mathematics skills are adding, subtracting, multiplying, and dividing whole numbers, fractions, and decimals. Skills can also exist as a domain. Take spelling, for example. Identifying a domain of words that a learner must correctly spell is easy. The same is true in mathematics. Any achievement test of skills should involve some representative sampling from the domain of skills.

An *ability* is a main interest of cognitive psychologists and many measurement theorists (Lohman, 1993; Messick, 1984; Mislevy, 2006; Sternberg, 1999). Cognitive abilities include reading, writing, speaking, and listening. Each cognitive ability is learned, but the learning process is very long and involved, perhaps spanning an entire lifetime. A cognitive ability requires the application of knowledge and skills in complex ways. Each cognitive ability has an emotional component as well (Snow & Lohman, 1989). Any cognitive ability is hard to develop and challenging to measure. Nonetheless, many cognitive abilities are the most important educational outcome for any educational program.

Intelligence is not an important theme in this book, because item writing, as featured in this book, is focused on tests that measure achievement. Intelligence has three constituent parts: verbal, quantitative, and analytical abilities (Haladyna, 2002a). These three abilities are more complex than the developing abilities that we consider part of achievement. Because of these three abilities' complexity, each ability is stable over a lifetime and not as easily modified as cognitive abilities like reading, writing, speaking, and listening.

Table 1.1 provides a hierarchy of cognitive traits. The ones at the bottom of the hierarchy are quickly and easily learned because of their low cognitive demand, while other types of learning are slow-growing largely due to their high cognitive demand. The three abilities that comprise human intelligence are the least resistant to change over a lifetime.

Table 1.1 A Continuum of Cognitive Behavior

Intelligence: Verbal, Quantitative, Analytical
↕
Developing, fluid, learned abilities
↕
Skills: Simple cognitive or psychomotor acts
↕
Knowledge: Understanding of facts, concepts, principles, and procedures
↕
Knowledge: Recall of facts, concepts, principles, and procedures

In light of Table 1.1, some subtleties exist that are useful later when observing how test takers respond to test items. If we have a change in cognitive behavior that we can attribute to teaching or training, then we might infer that achievement has occurred. Factors that are not relevant to teaching/learning may also account for changes such as cheating. Items will reflect this change due to learning in a pre-to-post comparison. If a student lacks an instructional history for some domain of content or some ability, then lack of instruction is the inference to make regarding the item's performance. That is, the item will perform as if it is invalid, when, in fact, the testing situation is inappropriate. If a student has received reasonable instruction and fails to perform as anticipated or hoped for, then something else has to account for that level of performance. What is probably accounting for test performance is not achievement but intelligence or lack of motivation to respond. Thus, the role of instruction or training and instructional history is an important consideration in deciding whether a test or test item reflects achievement or intelligence.

Validity

The most important concern in this book and for any test score or test item response interpretation is validity. Throughout this book and in each chapter, the essential concepts and principles

underlying validity are woven into presentations. Test items are developed for use in tests, but every item is also subject to an evaluation of its validity. As we will see, the development of test items is an integral part of an argument for item validity. This idea will be made clearer in the next section of this chapter.

The *Standards for Educational and Psychological Testing* (1999) state that validity is "*the degree to which evidence and theory support the interpretations of test scores entailed by proposed uses of tests*" (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999, p. 9). We will refer many times throughout the book to this resource, hereinafter referred to as the testing *Standards* (AERA, APA, & NCME, 1999). What process do we follow to enable us to assess the validity of a test score interpretation or use, or, in this book, of an item response?

In this chapter, we will employ an argument-based approach to validity advocated by Kane (1992; 2002; 2006a, 2006b) and, recently, successfully applied in a national testing program (Chappelle, Enright, & Jamieson, 2010). This argument-based approach accomplishes validation without a necessity for construct definition, which has been problematic in the past. The argument-based approach is not without precedent. Theorists have argued that validity should be conceptualized in forms of plausible arguments and challenges to the arguments (Cronbach, 1971, 1988; Messick, 1989). The idea of construct validation is based on inferences and assumptions presented in a logical framework.

Cronbach and Meehl (1955) provided three steps in testing: the definition of the construct, the explication of the construct—this means test development—and the validation. Kane (2006b) prefers a two-stage process: the development stage and the appraisal stage. With the argument-based approach we will highlight some major features of validity and validation as it currently exists, and in the final section of this chapter apply it to item development and validation.

Two Types of Constructs

The idea of two ways to approach measuring a construct can be traced to Messick (1994). Each construct type has a logical rationale and a set of procedures that enable validation. Both include domains from which a test is a representative sample.

The first of these two constructs is more traditional. It is a domain of knowledge and skills that is usually represented by a set of instructional objectives. Messick uses the term *construct-referenced*. A visit to any state education department website will reveal a set of instructional objectives organized by subject matter domains. These domains are the targets of instruction. Not very often, the cognitive demand of knowledge and skills is defined and used. Commonly, some tasks to be performed are more complex than knowledge and skills and resemble something else—a cognitive ability. This kind of domain has its origin in behaviorism and the criterion-referenced test movement so notable in the 1970s and later (Roid & Haladyna, 1982).

The second, less traditional, is a domain of tasks representing a developing cognitive ability, such as reading or writing. Messick uses the term *task-driven* for this type of construct. Writing SMEs will assert that the only way to measure writing is to obtain a sample of one's writing performance. Therefore, a domain of writing tasks might comprise the basis for testing of one's writing ability. The distinction between these two types of constructs is treated more fully in chapter 3. The main idea presented here is that the second type of construct is emerging due to the efforts of cognitive psychologists and others who are promoting *authentic assessment.* This second type of construct is preferred to the first type because it focuses on complex learning that is often overlooked with the use of the first type of construct. Both Messick (1994) and Lane and Stone (2006) articulate the rationale and progress needed to promote the valid use of test scores from this second type of construct.

Target Domain

Whether the construct is conceptualized as a domain of knowledge and skills or a domain of tasks representing a cognitive ability, we have a hypothetical domain of tasks to be performed. A simple example of a domain is what a first-year dental student must learn—the universal coding system for teeth. We have 32 teeth in the human dentition and 20 in the primary dentition. Given the tooth number or letter, the dental student must name the tooth. Given the name of the tooth, the dental student must give the identifying number. With 52 teeth and two variations, the domain has 104 different behaviors. This domain is a very simple target domain. A test of this target domain might entail a test of 20- or 30-item test that adequately samples from this domain. A target domain for dentistry is quite complex but readily identifiable. For instance, consider the 63 competencies required in a domain of knowledge and skills for dental licensing (Kramer & Neumann, 2003). This is only one part of a three-part licensing requirement. These competencies are complicated by the fact that each dental patient differs in many ways, such as age, patient problem, complications, and emergency situations. Table 1.2 shows the examples of competencies for patient treatment in dentistry. The target domain for dentistry is organized around the following major competencies: practice and profession, patient management, diagnosis and treatment planning, treatment.

So a target domain contains many tasks to be performed that should be grouped or organized using a taxonomy. These tasks cannot be realistically performed in a testing situation. As we can see, the term *target* is appropriate for helping us understand the meaning of the construct. A target domain is a heuristic device. Although the target domain may not provide realistic tasks to test, organizing it into a framework of content and cognitive demand that best reflects the collective judgments of SMEs is important. This framework should be our set of item and test specifications. In this document is a test blueprint or two-way grid, which will guide us in test design.

Although the target domain can be abstract in nature, hypothesizing that a target score exists is useful. It would be the score received if a student or candidate for certification were administered all of the tasks in this domain. In the example of writing, the target score would entail a lifetime of writing assignments assembled and scored accurately to achieve the target score—quite unrealistic. In dentistry, similarly, a candidate for licensure would have to show proficiency in all 63 competencies to achieve a target score. This is possible but very impractical. The target score has some implications for validity as we move to the universe of generalization (see Table 1.2).

Table 1.2 Target Domain and Universe of Generalization for Example Professional Competencies in Dentistry

38. Anticipate, diagnose, and provide initial treatment and follow-up management for medical emergencies that may occur during dental treatment.
39. Perform basic cardiac life support.
40. Recognize and manage acute pain, hemorrhage, trauma, and infection of the orofacial complex.
41. Manage patients with pain and anxiety by the use of non-pharmacological methods.
42. Select and administer or prescribe pharmacological agents in the treatment of dental patients.
43. Anticipate, prevent, and manage complications arising from the use of therapeutic and pharmacological agents employed in patient care.
44. Provide patient education to maximize oral health.
45. Manage preventive oral health procedures.

Source: http://www.jdentaled.org/content/68/7/742.full.pdf

Universe of Generalization

Realistically, with the help of its SMEs, a test developer can create a domain of tasks that can be performed by test takers practically in a testing situation. For educational achievement testing

programs as proposed in any state's content standards, these tasks should have considerably close connection to the target domain.

Fidelity is the closeness of any task in the universe of generalization to the target domain (Loevinger, 1957). In other words, fidelity is the resemblance of tasks in the universe of generalization to the tasks in the target domain. The judgment of fidelity is best made by SMEs. We have a very well-developed technology for matching tasks in the universe of generalization to the target domain. See Raymond and Neustel (2006) for examples dealing with professional competence type tests. See Webb (2006) for examples dealing with school achievement test content.

Another important characteristic of this universe of generalization is its organizational scheme. It should resemble or be identical to the target domain. For example, recently a research study on states' writing testing programs necessitated the development of a taxonomy of prompt types (Jeffery, 2009). Table 1.3 shows several tasks from a hypothetical target domain, and on the right side of the table a taxonomy of prompt types is given as an organizing structure for test development. Chapter 13 provides more discussion on how writing might be conceived in the framework of argument-based validity.

Table 1.3 Target Domain and Universe of Generalization for Writing

Target Domain Tasks	Universe of Generalization Framework
Writing a text message on your cell phone	Persuasive prompts
Making a sign	Argumentative prompts
Preparing a written speech	Narrative prompts
Giving written directions to someone	Explanatory prompts
Writing a textbook	Informative prompts
Writing a letter to the editor of a newspaper	Analytic prompts
Answering a survey	
Writing a personal letter	
Doing a movie review	
Preparing a report for a committee	

Referring to Table 1.2, the target domain consisting of 63 dental competencies is also the basis for the universe of generalization. By this fact, the fidelity of subsequent test specifications and tests is very high. The challenge to test developers is to create a universe of generalization consisting of testable tasks that accurately reflect these 63 competencies.

As with the concept of the target score, we have a universe score. This score is one achieved by a targeted test taker who is administered all items in the universe of generalization. It is what we know in classical test theory as the *true score*. However, the true score is not really true unless the target domain and target score have a great deal of fidelity with the universe of generalization and the universe score. Table 1.4 simplifies this discussion into the constituent elements in this content-related validity evidence assumption.

Table 1.4 Validity Features as Connected by the Target, Universe, and Sample

Validity feature	Target	Universe	Sample
Task score	Hypothetical set of tasks	Testable Domain of tasks	Test
Domain definition	Target Domain	Universe of Generalization	Sample from the universe of generalization
Score type	Target Score	Universe Score	Test score

With any test score, we have several inferences affecting validity. The test score reflects adequately the universe of generalization, and the universe of generalization has high fidelity with the target domain. In effect, the target score, universe score, and test score should be perfectly correlated.

Interpretive Argument and the Validity Argument

The basis for establishing validity via the process known as validation entails the *interpretive argument* and its partner, the *validity argument*. The interpretive argument starts with a description of the proposed interpretation and uses. This argument contains inferences and assumptions that support validity. Sometimes, these assumptions may be conditions that should be met or satisfied. According to Kane (2006a) we have at least three good reasons for having an interpretive argument:

1. A framework for test/item development supplies the assumptions and issues to be resolved.
2. The interpretive argument essentially is the validity argument sans validity evidence.
3. When we assemble validity evidence, the interpretive argument provides a basis for evaluating the validity argument.

In effect, the interpretive argument is the blueprint for gathering validity evidence that addresses questions, assumptions, and issues that affect the assessment of validity. The validity argument supplies answers to the questions that are subject to an assessment by a validator—a critical judgment about the degree of validity. Because validation is an exercise designed to confirm validity, this approach supports validity.

However, Cronbach (1988) and other validity theorists have also argued that we should examine threats to validity and weak links in the chain of inferences (Crooks, Kane, & Cohen, 1996). For instance, if all indicators for validity are sound, but reliability is very low, the validation fails. We have several good reasons for seeking evidence that may be disconfirming of validity. First, validation is a search for truth. One needs to examine forces supporting and refuting validity. Second, discovering threats to validity can only improve validity because subsequent remedial action can eliminate or reduce each threat to validity. So those engaged in validation should always seek validity evidence that might undermine validity. By doing so, a great service is afforded any testing program. In this book, the focus of validity and validation is both with test scores and item responses, simply because we interpret and use item responses just as we interpret and use test scores. Because items and item responses are subunits of tests and test scores, validity is also important for both item responses and test scores. The validity evidence we gather to support interpreting an item response is also part of the validity evidence we use to support the interpretation of a test score. However, some validity evidence may support a counterargument—that validity is not attained. Toward the end of looking for evidence that may undermine validity, Messick (1989) identified two major sources of threats to validity that should be considered.

Construct Representation: A Threat to Validity

As we consider the likeness of the target domain to the universe of generalization, as judged by our SMEs, it is important to note a major threat to validity that Messick (1989) identified. If we have a construct definition, we can discover how well the target domain represents this definition. If the target domain does not match the construct definition, we have a serious validity problem that falls in the category of content-related validity evidence. We argue that this kind of validity

evidence is fundamental to the meaning of a test score. The target domain may (a) misrepresent (b) overrepresent, or (c) underrepresent the construct. The difference in faithful representation is based on the collective, consensus judgment of SMEs.

First, evidence should be assembled showing that SMEs have developed an adequate target domain. A reconciliation is needed for the fidelity of the target domain to the fundamental qualities of the construct. For instance, with writing, if we had a domain of writing tasks comprising our target domain, how closely does this domain resemble writing as it exists in our society? Second, evidence should be assembled showing that SMEs have developed an adequate organization or structure for tasks. The tasks in the target domain should be arranged in a taxonomy, but the job is not finished. When the universe of generalization is developed displaying the test tasks, the SMEs must determine how much correspondence exists between the content of the target domain and the content of the universe of generalization. In the instances of writing as shown in Table 1.3, what percentage of representation does each prompt mode have in the assessment of any student's writing? Documentation of correspondence of test content to construct content is critical. To put this in the language of the argument-based approach and current view of validity, the target domain reflects the criterion tasks to be performed. Given that such performance testing is unrealistic, the universe of generalization assumes the ability to simulate the tasks in the target domain. The fidelity of tasks in the universe of generalization and the target domain is a critical feature of construct representation.

Construct-Irrelevant Variance: Another Threat to Validity

According to Messick (1989), a second major threat to validity is construct-irrelevant variance (CIV). With reliability, we have random error, which can be large or small, positive or negative. We can think of CIV as systematic error that either inflates or deflates a test score. This kind of unwanted influence is also commonly called *bias*. Haladyna and Downing (2004) defined CIV and provided many documented instances of CIV. Some common examples include cheating, scoring errors, fatigue, rater severity, and extended time limits for some test takers. CIV can be studied, evaluated, and reduced or eliminated. In the study of test items, we identify many sources of CIV with item responses. In particular chapters 16, 17, 18, and 19 deal with most instances of CIV involving items.

Validation

As we know, validation is an investigative process that has several important steps: the development of our interpretive argument and the development of the validity argument. The test provides a test score that is an estimate of the universe score. The test score, and universe score, and target score should be perfectly correlated if our content-related validity evidence for the test is in order. Thus, we assemble evidence to support many issues that affect the interpretation and use of test scores, and, in this volume, item responses. The assembling of evidence is very systematic. The body of evidence must be assessed to support the claim for validity or the evidence may have weak links that cast doubt about validity.

Where Do We Go From Here?

The previous discussion about validity has been abstract with a few concrete examples in dentistry and in writing. In the final section of this chapter, we apply the argument-based approach to the important process of validating item responses. We call this process *item validation*. A major step in test development is item development. Item quality is a chief concern. Item validity is a main link in the chain of inferences in test score validation. By focusing attention on item validation we satisfy a critical step in test development.

In this final section of this chapter we apply the concepts and principles of validation discussed previously to item validation. For an item response to be valid, we have a set of assumptions as questions. If these condition represented by each condition can be documented as having been met, both the interpretive and validity arguments for item validation can be satisfied.

An Argument-Based Approach to Item Validation

The balance of this chapter will present each question and consider what kind of evidence is needed to satisfy this condition for item development and item validation. Table 1.5 provides an overview of the questions to be answered. Downing and Haladyna (1997) argued that item validation involves many important steps, each of which constitutes a different and complementary type of validity evidence. Each question represents one type of evidence that bears on item validity. Failure to satisfy the condition underlying the question is a weakness in the chain of inference supporting validity. Often, other chapters and references to others' work provide more comprehensive treatment of the types of evidence needed and how to obtain it.

Table 1.5 Interpretive Argument for Item Validation

1. What type of target domain is intended?
2. How is the target domain organized?
3. How is the universe of generalization organized?
4. How much fidelity is there between the target domain and the universe of generalization?
5. Which item formats will be used?
6. How are items developed?
7. Is the scoring key accurate?
8. What is the content for each item?
9. What is the intended cognitive demand for each item for a typical set of test takers?
10. Were items edited?
11. Were items reviewed for currency and effectiveness?
12. Were items reviewed for unnecessary linguistic complexity?
13. Were items reviewed for fairness?
14. Were items pretested effectively?
15. Was the internal structure of the test content studied?
16. Who decides whether an item stays or goes or gets revised?

1. What type of target domain is intended?

As noted previously, a target domain can consist of (a) knowledge and skills or (b) a set of tasks that require the use of knowledge and skills. Most state content standards for language arts, mathematics, social studies, and science appear to focus on the former type of domain, but some applications seem more suited for the second type of domain. Writing comes to mind as a domain of tasks of the second type. In professional testing for competence, most domains are an organized set of performed tasks—the criterion behaviors of a professional practice. However, many certification and licensing boards continue to also measure the domain of knowledge and skills that are essential to professional practice.

2. How is the target domain organized?

For a target domain to have adequate meaning for SMEs and consumers of test scores, the domain of knowledge and skills or the domain of tasks should be organized as a hierarchy. As noted

previously, the 63 competencies for dentistry are organized in a hierarchy: practice and profession, patient management, diagnosis and treatment planning, and treatment (Kramer & Neumann, 2003). Writing can be thought of as existing in six distinct prompt modes. Chapter 3 provides more information about this type of item validity information. Kane (2006a) discusses content-related validity evidence as a major concern in test development. Organizing target domains is a vital step in achieving a body of content-related validity evidence.

3. How is the universe of generalization organized?

Because this domain consists of test items representing tasks that comprise the universe of observation, we can imagine a bank of test items that adequately represents this universe. The device that is widely recommended for this organization is the set of item and test specifications. Again, this type of evidence is content-related. Good sources for developing content-related validity evidence include Raymond and Neustel (2006) for the professions and Webb (2006) for educational constructs. Drawing from dentistry, the organization of competencies in the target domain is the same as it is for the universe of generalization. For the writing example, we have a proposed taxonomy of prompt modes from which to represent content (Haladyna & Olsen, submitted for publication). Chapter 3 provides more information about this type of item validity evidence.

4. How much fidelity is there between the target domain and the universe of generalization?

With item validation, we seek the consensus judgment of SMEs that the universe of generalization that contains our item bank has high fidelity with the tasks in the target domain. We are very concerned with the degree of judged fidelity. As noted previously, with the dentistry competencies, fidelity is very good. With the writing domain, the fidelity of the six prompt modes to the set of writing tasks employed in our society is unknown. The proposed taxonomy of prompt modes may reflect common practices in the United States, but there is no reference to a target domain of writing tasks widely practiced throughout this nation. Chapter 3 presents more information about this type of item validity evidence.

5. Which item formats will be used?

Given adequate information about the cognitive demand of different item formats and the item formats best suited for types of content, the SMEs will judge the item formats to be used. This judgment should be based on sound, stated principles that reflect best practices. Chapter 4 provides information about the choice of item formats. Formats differ in the amount or degree of content-related fidelity but other trade-offs may justify the selection of lower-fidelity item formats for great efficiency or reliability.

6. How are items developed?

This book helps to develop test items that will pass the test of item validation. Chapters 5 through 15 provide very specific information about item development. We have many important steps in item development. Some of these steps are (a) the development of an item-writing guide, (b) the development of an editorial or style guide, (c) selection of highly qualified SMEs who will write items, (d) procedures for many reviews mentioned in this section and covered more adequately in chapter 16, and (e) training of these item writers. The *Handbook of Test Development* (Downing & Haladyna, 2006) has many chapters devoted to item development. Also, Haladyna (2004) has an extensive treatment of SR item development. The balance of this book contains many recommendations for item development and validation.

7. Is the scoring key accurate?

With any SR item, the correct answer must be absolutely correct. A consensus is required from SMEs. This is not a vote with a majority. Agreement should be perfect. With CR items, if an item is scored dichotomously, again a consensus about correctness is essential. With the CR format, several correct answers might exist. If a descriptive rating scale is used, the development of the rating scale is a very important activity, which is treated in chapter 12. With human judgment comes a host of threats to validity that include CIV and rater inconsistency. Some content issues arise with the development of the scoring guide (rubric) for rated performance, which are discussed in chapter 12 and more thoroughly in chapter 18.

8. What is the content for each item?

Each item must be accurately classified according to the set of item and test specifications. This activity of the SMEs should be well documented in a written, dated report. The report should show that every item in a bank has an accurate content designation that is compatible with the designations in the test specifications. When the test is designed, the content categories from the test specifications must be satisfied. Chapter 3 discusses content issues.

9. What is the intended cognitive demand for each item?

Each item should be judged for its cognitive demand for typical test takers. Chapter 3 discusses the cognitive demand. The importance of this step is high because of the difficulty we have in conceptualizing cognitive demand and the fact that test takers vary in the cognitive demand actually employed in responding to any item. Nonetheless, the pursuit of identifying cognitive demand for targeted test takers is an important goal that affects how well an ability is developed and how validly it is measured. There is a persistent and urgent need for test content and test items to have a higher cognitive demand (Lane, 2004; Nichols & Sugrue, 1999). Despite the shortcoming of methods for classifying cognitive demand, greater emphasis in item and test development is needed here. One important source of validity evidence recognized by the testing standards is response processes. One method of investigating response processes is the think-aloud or cognitive interview, which also can be used to verify the cognitive demand of test items.

10. Were items edited?

The benefits of editing are many: improve the clarity of items, correct spelling and punctuation errors, correct grammar, ensure that vocabulary is appropriate and acronyms are qualified, ensure that the reading level is appropriate for the test takers, ensure that the item is presented clearly to the test taker, and carry out final proofing of a test. Each item should be edited using an editorial style guide. A chapter by Baranowski (2006) on this topic is very informative about editing. Professional editors are highly desirable in any testing program. However, the concepts, principles, and procedures in editing are often performed by other personnel. For small testing programs or classroom testing, the item writer is often the editor as well.

11. Were items reviewed for currency and effectiveness?

This review is seldom a problem for scholastic achievement, because the content of most scholastic abilities remains constant over many years. Reading, writing, speaking, and listening has not changed that much. Mathematics content is very stable. Although science is an actively changing field, principles of science are fixed due to the laws of nature. However, as professions change often, currency is a critical concern. With new technologies, new legislation, changes in the population demographics, and other factors make items for professional certification and licensure

extinct or useless and require new items. Therefore, a periodic review of items by SMEs is necessary to refresh the item pool.

12. Were items reviewed for unnecessary linguistic complexity?

A considerable source of CIV in any testing program is linguistic complexity that is not part of the construct being measured. In educational achievement testing, instead of testing language proficiency, tests should primarily measure the content intended. Research by Abedi (2006) and research he cites show that unnecessary linguistic complexity can result in an underestimation of a student's ability due to the student's weaker language proficiency. Some modifications to improve clarity involve word frequency and familiarity, word length, sentence length, voice of verb phrase, negation, and sentence structure. Abedi offers many other factors that may lower test performance inappropriately, and he offers procedures for dealing with this problem. Although chapter 16 provides a brief treatment of this problem, his chapter in the *Handbook of Test Development* is highly recommended.

13. Were items reviewed for fairness?

As with other CIV factors, fairness review addresses a source of CIV that may affect only several test takers. Nonetheless, fairness review is important enough to be undertaken in most testing programs and is a fixture where the public's welfare is concerned. The Educational Testing Service has been a leader in this field, and provides the *ETS International Principles for Fairness Review of Assessments* (2004; available at www.ets.org). A good source of information about fairness review can be found in Zieky (2006). Chapter 16 provides a discussion of fairness review and recommends that one be done for all item development efforts.

14. Were items pretested effectively?

New items should not be used to comprise the total test score. New items should be pretested to assess their difficulty and discrimination. Field-testing of new items has many guiding principles and recommended procedures including the use of varying test theories. A record of the result of pretesting needs to be documented without violating privacy or exposing items. Chapter 17 provides a comprehensive treatment of statistical study of SR test items. Chapter 18 provides complementary information on CR test items. Chapter 19 deals with special problems with test item responses.

15. Was the internal structure of the test content studied?

It matters greatly whether a test's content is unidimensional or multidimensional. Knowing the structure of data helps estimate reliability more accurately. Also, the structure of data may enable or prevent the responsible reporting of validated subscores. A validity study of internal structure should be done and reported in an annual technical report or other special report. Chapter 17 discusses the issue of dimensionality. Other references cited in that chapter provide more extensive information about the study of dimensionality and its effects on validity and reliability.

16. Who decides whether an item stays or goes or gets revised?

A persistent conflict in item development is who ultimately decides whether an item should be revised or retired or stay active in the item bank. On the psychometric side, the argument is if the difficulty and discrimination of the item are favorable; such an item when used in a test contributes to reliability. As reliability is a very important type of validity evidence, who decides the fate of a test item? In other words, do psychometric criteria trump all other factors?

The opposing argument is that the panel of SMEs is the ultimate authority regarding each item's appropriateness. An easy item is answered correctly by more than 90% of all test takers.

Although the item may appear to be too easy, the item may represent important content expected to be learned. Thus, the SME wants the item to stay in the test. Or even though an item is too difficult and fails to discriminate, it stays because it tests something important that is not taught very well or is not learned by candidates for licensure. We support the judgment of the SME panel, but with the proviso that psychometric criteria are revealed to this panel and considered. Using solely psychometric criteria to decide the future of the item may improve reliability but at the expense of content-related validity evidence.

Technical Documentation

This chapter has focused on the need to build an interpretive argument and satisfy a validity argument for item validity. A primary device for documenting this validity evidence is the technical report for any testing program. Thus, it is strongly recommended that the periodic technical report provide all the validity evidence suggested in Table 1.5. This advice is consistent with many testing experts who have consistently advocated documenting validity evidence responsibly (Becker & Pomplun, 2006; Ferrara, 2006; Haladyna, 2002b).

Summary

In this first chapter, test item and test were defined. A major theme throughout this chapter has been the role that validity plays in making test score interpretations and uses as truthful as possible. As test items are the essential building blocks of a test, validating item response interpretations and uses is just as appropriate as validating test score interpretations and uses. An argument-based approach is used. Many propositions were provided that comprise essential steps in item development. These 16 questions support an interpretive argument for item validation. The validity argument requires that evidence be collected and organized. Such evidence might appear in a report concerning item quality or as part of a technical report. As we document these steps, validity evidence is displayed in our technical report, and, by that, validity is improved. Weaknesses in item validation can be very deleterious to test score validity, which is why considerable attention should be given to test score validation.

2
Developing the Test Item

Overview

This chapter presents the essential steps required to develop any test item. The chapter is intended for those who are interested in developing a validated item bank for a testing program. Whereas subsequent chapters deal with developing items for different types of item formats, this chapter is intended to outline the steps involved in developing any test item.

Planning

At the beginning of item development for a new testing program, the item bank will be empty. At this point, a plan should be created for filling the bank. Even for existing testing programs with extant items, a plan is always useful. The plan will include many procedures and issues, which include the number of items needed, the types of item formats to be used, the rationale for using these item formats, the type of content and cognitive demand intended, the personnel responsible for item development and validation, and, most important, a schedule.

For selected-response (SR) items, a rule of thumb is that the item bank should be 2.5 times the size of a test. This number is purely a subjective value that is based on opinion and experience. For a 100-item test, a reasonable goal is to validate 250 items that match your needs as specified in your item and test specifications. If multiple test forms are being used annually, the number of items in the bank should be proportionately higher.

For constructed-response (CR) items, the determination of how many items are needed is very difficult to ascertain because CR tests vary considerably in the number of items. This number will be governed by the type of test for which items are being developed. Many subsequent chapters are devoted to developing CR items with unique qualities, so better guidance is found in those chapters.

Because many tests appear in a SR format, we will limit discussion to this format, but the steps apply equally to CR formats as well.

The inventory is a private planning document shared between test developers and the test sponsor. The inventory resembles the item and test specifications (which includes the test blueprint) but shows the supply of validated items available for test construction. The inventory will also reveal what content and cognitive demand categories are deficient with respect to the desired levels of items in each cell of the test specifications.

Table 2.1 shows a test blueprint (also known as a *two-way grid*) for topics and cognitive demand that shows the ideal inventory for a hypothetical item bank. This is a 100-item imaginary certification test for professional meal planners.

Table 2.1 Ideal Number of Items in the Item Bank

Topics	% Coverage of topics	Knowledge	Skill	Ability	Desired (by topic)
Basic principles	20%	15	15	20	50
Planning	25%	19	19	25	63
Preparation	20%	15	15	20	50
Presentation	20%	15	15	20	50
Clean-up	15%	11	11	15	37
Desired (by cognitive demand)	100%	30%	30%	40%	250 (100%)

As shown in Table 2.1, an ideal number of validated items in this bank is 250. The number of items in each cell shows the proportion of items desired. The percentages show the emphasis for each topic and cognitive demand. These percentages were recommended by subject-matter experts (SMEs) who examined the results of a practice analysis.

Table 2.2 shows the actual number of items in the item bank, based on the structure of the same test blueprint (two-way table). For the first topic, basic principles, we only have 40 items and need 10 more. However, we need three more knowledge items and nine more skill items, and we have an excess of ability items (which require the use of knowledge and skills).

Table 2.2 Actual Number of Items in Item Bank

Topics	% Coverage	Knowledge	Skill	Ability	Total
Basic principles	20%	12	6	22	40
Planning	25%	50	12	28	90
Preparation	20%	11	22	10	43
Presentation	20%	13	11	31	55
Clean-up	15%	9	12	9	30
Total	100%	95	63	100	258

A periodic inventory is easy to complete and provides a very accurate account of how many validated items are available for test design. Moreover, in planning for future item development, assignments to SMEs should be made with the inventory results in mind. That is, if a meal preparation expert is especially highly qualified in meal presentation, that specialist should be assigned to write items in that category. As Tables 2.1 and 2.2 show, knowledge and skill items are needed, and no ability items are needed. In planning for future item development and validation, these two tables provide excellent guidance.

Item Bank

One aspect of planning is to decide where test items are kept. The item bank is a modern, indispensable tool for housing test items and associated information about the performance of the item with test takers. The most informative and complete discussion of item banking can be found in Vale (2006). The history and technology supporting computerized item banking is both interesting and useful. However, item banking is a developing technology and new and more effective products are constantly being released for public use. Many test companies have proprietary item banking systems that are not available to the public.

The item bank has two major functions. One, it should keep the validated item in camera-ready format ready for placement on the test. Two, the item bank should contain a history of the item including its difficulty (p-value), discrimination index, and other relevant information. If the item is in a SR format, it should have a frequency of response for each option. If performance and a rating scale are used to score a response, it should have a frequency for each rating category. If more sophisticated scaling methods are used, such as those involving item response theory, appropriate statistics from the various item response theory models should also be recorded in the item bank.

Of particular interest is a typology of items that Vale (2006) has introduced. He conceives of test items in a social order. The value of using a social order is to avoid the placement of items on a test in inappropriate ways that may threaten validity.

- *Friends* are items that must appear together because of some similarity. For example, in a subsection of a mathematics test, we might enjoy presenting items on geometry as a set.
- *Close friends*. In some item formats, items must be grouped together due to the dependence on a stimulus, such as in reading comprehension where a passage is presented before the test items. This type of format has been referred to as an item set or testlet (Haladyna, 2004).
- *Snobs*. Some items must appear in a specific way in proximity to other items. There are no exceptions with snob items. Vale uses the example of punctuation in a series of sentences comprising a paragraph. If the items were reordered and did not conform to the order of the example, the items would confuse test takers.
- *Dependents*. Some items need supporting material. A typical example would be the item set, where each item cannot be answered unless a passage, vignette, photograph, or another stimulus precedes the set of items. These items are truly dependent.
- *Supporters*. These items have no interaction with other items but support other items. We might conceptualize supporters to be critical features of an item set on problem-solving where one item is NOT a cue for another item.
- *Antagonists*. Any item that presents a cue for another item is an antagonist. Chapter 6 presents information on different types of cuing for SR items.
- *Enemies* are simply items that cannot appear on the same test. These might be items testing the exact content.

Table 2.3 lists properties of items that an item bank might contain. The decision of which properties are appropriate is based on many factors, such as type of test, types of test items, varieties of item formats, and type of scaling use (featuring item response theory or other methods).

As noted previously in this chapter, software for item banking is challenging. We have commercially available products and test companies have proprietary item banking software that is not publicly available. If you have contracted with a test company, each company can describe how they bank items and their item-banking capabilities. If you have to bank your items without the benefit of a test company, your only option is to purchase one of several commercially available item banking systems or create your own homemade system. Table 2.4 provides a list of commercially available item-banking systems available. A homemade item banking system would use a word processing program such as WordPerfect or Word for the camera-ready image of the item. A spreadsheet can be used for recording item history with a common identification code to connect the word processing file with the spreadsheet. A proviso about commercially item-banking software is that these products have a dynamic nature and are continually changing. Also, with the introduction of new item formats as shown in different chapters of this book, competing software will have the same capabilities. So, as a potential user of any software system, matching up the needs of the testing program with the software is important.

Table 2.3 Properties of Item Banks

Property	Meaning
Ancestry	When the item was written, when it was revised.
Content classification	Contains the content code or some finer delineation consistent with item and test specifications.
Identifier	An identification code. This code might include the content code as well.
Item difficulty	Presented in percentage correct or using an item response theory metric.
Item discrimination	Presented in a classical sense as a discrimination index or using the item response theory metric.
Answer justification	A discussion of why the correct answer is correct and why other answers are not correct. If a constructed-response answer, some guidelines should be given for the range of answers if a rating scale is used. If the constructed-response is a short answer, the list of acceptable short answers should be given and approved by the SMEs.
Quality rating	In some circumstances, items should be rated by reviewers regarding the item's usefulness in future examinations (e.g., excellent, good, fair, poor—revise or retire).
Response type	Selected-response or constructed-response.
Reviewers	As all items are reviewed, a history of reviews and results of review.
Revision history	A history of all revisions might be included. The item is shown as originally written and in all other versions including the current version.
Scoring key	The correct answer should be given for a selected-response item. For some constructed-response item types, the correct answer (or set of correct answers should be provided). For constructed-response items where a rating scale is used, benchmark answers representing various points on the rating scale should be given.
Social order	As just presented.
Source reference	For scholastic achievement, textbooks are a good source reference; for professional testing we use journal articles or textbooks.
Usage history	A record of when the item was used and the results of that use.
Writer	Who wrote the item.

Adapted from Vale (2006).

Another possibility for item banking is any relational database, such as Access, widely disseminated with Microsoft Office or similar office packages. Relational databases allow for the creating of multiple tables to be linked through a common identifier. For example, a test item can be created with a unique identification and, with that identification, the item can exist in multiple tables, including a table that contains the item itself, a table containing the history of the item development, a table containing associated graphics, a table containing item statistics, a table containing item content and cognitive demand information, etc. The tables are linked

Table 2.4 Examples of Popular Commercially Available Software Providing Comprehensive Testing Services That Include Item Banking

Name	Web Address
FastTEST 2.0	fasttestweb.com
FastTEST Pro 2.0	fasttestweb.com
FastTEST Web	fasttestweb.com
Perception	questionmark.com
Random Test Generator PRO	hirtlesoftware.com
Test Creator	centronsoftware.com
Test Generator	testshop.com
Testdesk	aditsoftware.com
Unitest System	sight2k.com

through the unique item identification. Any subset of information related to a specific item or set of items can be obtained through the development of specific queries (a summary table generated on demand to report on information from multiple tables for a specific item).

For more information about computerized item banking, the reader is directed to Vale's (2006) chapter 11 in the *Handbook of Test Development*.

Identifying and Recruiting Item Writers

SMEs are very important for item development. Their professional opinions about content and cognitive demand of each item and other activities performed by them are actually forms of validity evidence. Sometimes, the opinions of testing specialists may conflict with the opinions of SMEs. In matters of content, the SMEs' opinions should prevail. In such matters as reliability or scaling, the testing specialists' opinions are usually better. However, content-related validity evidence is a paramount type of validity evidence.

The selection of SMEs should be based on their qualifications and not just their willingness. The most highly qualified SMEs should be recruited. As part of the search process, it is a good idea to obtain their resume/vitae to document the high abilities of these SMEs. As they make important decisions throughout the item and test development process, this ensures that you have content experts working to validate your test items and test score interpretations and uses. The documentation of the search for and enlistment of highly qualified SMEs is a small but important piece of validity evidence.

Invariably you will discover that the most highly qualified SMEs may not necessarily have the best item-writing ability. Sometimes, you will find item writers who may lack the loftiest credentials but have a knack for writing really good items. Some SMEs may not learn how to write items or follow the guidelines. So, having highly qualified SMEs is necessary, but there is not necessarily a high correlation between expertise and quality of items. Nonetheless, item writers are SMEs, and they should have records of high status in the field for which they were chosen to write items.

Assignments to Item Writers

Based on the inventory of the item bank, and the results showing excesses and deficiencies, item-writing assignments need to be made. The unique expertise of each item writer/SME should be considered. Try to match the expertise of each item writer to your needs for new items. These item writers should also review previously validated items, as called for in your plan. It is useful to let them know how many items they are expected to write and how many items they are expected to review. A schedule should be established, and item writers need to be reminded to follow the schedule. Constant monitoring of item writers is necessary.

It is also practicable to have item writers write items in teams or at meetings where interaction and feedback are sought. The item writers create items on the fly and the items are reviewed by other SMEs. Sometimes in a round table or a group meeting, obtaining expert commentary on each item can be a slow, tedious process, but the benefit is that each surviving item has the approval of most of your SMEs.

If item-writing is done privately and individually by the SME in the workplace or home, the item assignments are handled best via mail or email. The same level of review can be done very efficiently. Again, monitoring of item writers in all steps is essential to staying on schedule.

After the item has been written and approved by SMEs, the item goes through an assembly line of checks and reviews as briefly described in this chapter and more completely discussed in chapter 16.

Item-Writing Guide

An item-writing guide is the official document used in the training of item writers and used by item writers to help them in the item-writing process. The item-writing guide should contain the item and test specifications, some information about the inventory so that they can understand their role in writing items, formats to be used and not used, and item-writing guidelines, such as suggested by Haladyna and Downing (1989a, 1989b), Haladyna, Downing, and Rodriguez (2002) and in this book in chapter 6.

One of the most comprehensive item-writing guides can be found on the website of the National Board of Medical Examiners (http://www.nbme.org/publications). This item-writing guide is the longest you will encounter, but it shows detail and thoroughness are possible. On the other hand, SMEs are not likely to read these longer item-writing guides as thoroughly as you might want, so shorter item-writing guides might be more effective for getting SMEs started. (See Table 2.5.)

Table 2.5 Outline for an Item-Writing Guide

1.	Brief description of the testing program
2.	Description of the item classification system for content and cognitive demand
3.	Instructions on how to prepare and transfer items and other logistical concerns
4.	Item formats to be used and not used
5.	Examples of well-written and poorly written items
6.	Item-writing form/template (electronic versions preferred)
7.	Guidelines for writing items (DOs and DON'Ts)

Item-Writing Training

The training of item writers is an important event. Conducting this training constitutes evidence for item validation and for test score validation. Most untrained item writers have predetermined habits for writing test items that will not produce validated items. Most untrained item writers have almost no experience writing test items. Therefore, training provides each SME with an opportunity to develop knowledge, a set of skills, and some strategies that comprises their item-writing ability. The training uses the item-writing guide and begins with a didactic introduction. A typical session might include the outline provided in Table 2.6.

Table 2.6 Outline for Item-Writing Training Session

1.	Introduction and overview
2.	Definition of content/construct
3.	Structure of the test (item and test specifications)
4.	Formats we use
5.	Guidelines for writing items
6.	Examples of well-written and poorly written items with group discussion
7.	Forms to use when writing items (preferably using a word processing program)
8.	Item-writing session
9.	Post-session group discussion of results
10.	Editing and reviewing items
11.	Characteristics of validated items (including item analysis)
12.	Assignments and schedule

One of the most useful, valuable activities in item-writing training is the post-session group discussion. To hear colleagues discuss your item and offer constructive advice is valuable both for improving the item and for learning how to write better items. The length of the session can be as little as a few hours or can be extended to several days if the group of SMEs is on a production schedule. The editing of items is often done by a professional editor, but preliminary editing helps. The reviewing of items is an ongoing activity over several months.

Editorial and Style Guide

The editorial and style guide is a tool for the editor. It is usually a very brief document that contains guidelines, rules, and information about item formats, presentation of items, uses of slang or popular terms, conventions, the use of acronyms, special vocabulary, and other editorial concerns. Such guides are common place in business and education for written communication because it provides for standardization. For any test, this kind of standardization is crucial. Items must clearly communicate a cognitive task. The editorial and style guide is one part of this extensive effort to create validated test items.

Drafting the Test Items

After each SME has received an item-writing guide, training on item-writing, and a schedule for item-writing, the SME drafts new items. Some item writers have trouble getting started; they experience *writer's block*. A remedy to be found in good item-writing guides, and discussed in chapter 8, is the use of item shells. These devices are skeletal remains of previous items. Each shell provides part of a previously used validated test item. For existing testing programs, some items will invariably be retired or flagged for revision. Apart from the item shells, chapter 8 provides many schemes for item generation that represent shortcuts. A new item writer can engage in creating a new item based on the retired item by correcting the faults observed in the retired item. This new item writer can also revise items. Both activities give the novice item writer practice and experience.

Drafting test items is a very difficult activity. Very few SMEs are fluent in item-writing. Item-writing requires expertise in content and a focus on the specific content and cognitive demand required. Fortunately, item-writing is often a team effort with others being involved. The first draft of the item is just the beginning.

Providing the Key or Model Answer

If the item is in a SR format, the key (correct answer) is selected by the item writer. The key must be verified by at least one or more SMEs. If there is any disagreement, a team meeting should be held to decide correctness. If another option is thought to be a key, it might be modified or replaced. Correctness should never be a matter of voting. Correctness should be determined by 100% agreement.

If the item is in a CR format, the key or model answer can be a simple determination or a complex issue. It depends on the type of format. Chapter 4 discusses how formats vary within the SR and CR categories and issues of measurability of content. For our purposes, we need to address fundamentally different CR formats that depend on how the response is scored: subjectively or objectively. Chapter 12 directly takes on the task of scoring CR item responses.

CR: objectively scored. Some of these CR item formats require a simple word or sentence response, scored right/wrong, or require a performance of a simple skill. In both instances, a

scorer must determine if the item is correctly or incorrectly answered. Judgment is needed, unless the response is objectively determined.

CR: subjectively scored. One of the best examples of this type of item scoring is a writing prompt where a rubric (descriptive rating scale) is used. Chapters 10 and 11 provide specific information about these kinds of items and how to design these types of items.

The scoring of these items is subjective, and requires the judgment(s) of SMEs as described in chapter 12. Unlike objective scoring where there is only one answer or a set of agreed upon correct answers, the judgment is made in terms of degrees. To aid the SMEs in making judgments, benchmark performance examples might be provided in a training session to find out if the SME is performing in accurate and consistent ways.

Classifying the Content of Each Item

This critical type of item review is more completely discussed in chapter 3. Classifying the content of each item is a very important step and a piece of content-related validity evidence.

The basis for classifying items is found in item and test specifications. Chapter 3 provides more information about this document and its importance. Tables 2.1 and 2.2 show the organization of content for a hypothetical meal preparation certification test. With a few content categories, it might be good to provide a finer degree of content organization. For instance, for the presentation category, it might be desirable to add two categories, such as formal and informal. The finer discrimination is a device to prevent writing items that have too similar or identical content. By knowing that items are distributed by these finer distinctions, we can decide how a greater degree of balance can be achieved in filling our cells with validated items.

Estimating the Cognitive Demand for the Targeted Learner for Each Item

As chapter 3 is devoted to the topic of content and cognitive demand, the task of labeling each item with an expected cognitive demand gets extensive treatment there. As test developers continue to lament the lack of test items with the more desirable higher cognitive demand, we have a considerable challenge ahead. Each item should be given a designation for the intended cognitive complexity that the typical test taker has to undergo to select or construct the right answer. The system for organizing items by cognitive complexity is a matter of considerable concern as chapter 3 shows.

Providing an Authoritative Content Reference

As stated in the previous section, the key/correct answer for any item should be accurate. If a team of SMEs is used, the decision should be based on 100% agreement and not subject to a vote. Another way to ensure that the correct answer is really correct is to provide a reference from an authoritative text that is commonly accepted. For school achievement, textbooks are one source. However, the textbooks may have to be on an approved or accepted list. In professional fields, eminent textbooks provide authoritative information and journal articles are also valued. Sometimes, the authoritative source is a curriculum.

Reviewing the Test Items

Chapter 16 presents information on a series of reviews that comprise a body of validity evidence for item validation. As noted previously and elsewhere, reviews comprise a qualitative type of

validity evidence. Each type of review is complementary. A series of reviews is strongly recommended. Each intends to ward off a threat to validity. Each is independent of the others.

Fairness

Although fairness has been a concern of test developers and test users for many years, we have no widely accepted definition (AERA, APA, & NCME, 1999, p. 80). One definition that works for item validity is that any characteristics of items that affect test scores and are unrelated to what is being measured are unfair. That is, the item elicits construct-irrelevant variance, which is a major threat to validity (Haladyna & Downing, 2004).

One of the most significant efforts to date on fairness comes from the Educational Testing Service Fairness Review Guidelines and the efforts of Zieky (2006). Fairness review is highly recommended. First, guidelines provide a concrete basis for determining what is fair and unfair. Subjective judgments of fairness are less likely. If the guidelines are universally shared and published, we are less likely to have unsuitable content in tests. From his research of similar guidelines on fairness, Zieky lists six guidelines that one might consider:

1. Treat people with respect.
2. Minimize the effects of construct-irrelevant knowledge or skills.
3. Avoid material that is unnecessarily controversial, inflammatory, offensive or upsetting.
4. Use appropriate terminology to refer to people.
5. Avoid stereotypes.
6. Represent diversity in depictions of people.

Zieky also recommends various ways to adjudicate fairness reviewers' disagreements. Generally he favors a third party, whether it is a dispassionate expert or a large committee.

Aside from fairness review, we have other remedies that do not fit in this category. For instance, differential item functioning is a statistical technique that uncovers an empirical basis for unfairness—where an item provides an advantage for one group over another. Another major area involving fairness is accommodations. Chapter 15 deals with item development for students with exceptionalities. Chapter 16 provides more information about fairness.

Language Complexity

With growing awareness about how those learning the English language take an English language test, we see more research on the nature of language complexity in item development and how it might present instances of construct-irrelevant variance. As with fairness, the linguistic complexity of test items may lower a test taker's score unfairly.

The central issue is that if a test does not measure reading comprehension, we do not want reading comprehension to be a factor in determining the performance on the test. That is, a test is supposed to measure one construct alone and not many constructs. There is extensive and growing research that the degree of linguistic complexity does affect test performance (Abedi, 2006). By simplifying the complexity of test items in an appropriate way, test performance increases for some test takers. We think that such accommodation is fair, as reading comprehension should not interfere with the performance on a test. In theory, we think it is the cognitive demand of reading comprehension that influences test performance. Abedi's chapter provides the most extensive treatment of this subject to date.

What are some features of linguistic complexity that should concern us?

1. Word frequency and familiarity. Words high on the word frequency list are more likely to be read and understand than low frequency words.

2. Longer words have a documented high degree of difficulty.
3. Longer sentences also affect performance.
4. Passive voice is more difficult to understand than active voice.
5. Negation in the stem of any item should be avoided.
6. Concrete over abstract descriptions of problems produce higher scores.

Abedi lists many other factors that address language complexity. It is a growing science with proven results. A review of language complexity is strongly recommended, as it is a fairness issue. Abedi provides extensive examples of language simplification activities intended to improve clarity and measure the construct more validly.

Editorial

Depending upon the size of the testing program, editing is done professionally by a highly trained and well-qualified specialist or it is done informally by someone with good knowledge and skills of editing. In either situation, the goals of the editor are namely (a) revise items to improve clarity but NEVER change content, and (b) correct grammatical, spelling, punctuation, and capitalization errors. The editor also insures that the item is presented in the correct format, so that when the item goes into the item bank, it is ready for use in a test.

Another important activity of the editor is proofing. Although proofing may be a shared responsibility, the editor is best trained and qualified to proof. The standard for publication of any test is perfection.

A chapter by Baranowski (2006) provides much useful information about editing. She considers the editorial review as a type of qualitative validity evidence, a view that is consistent in this book as well. The editorial style guide is a useful document to this editor.

Field-Testing the Item

Once an item has been developed and gone through the many steps outlined in this chapter, the item has to be tested with test takers from the target population for which the test is intended. Two aspects of field-testing exist. The first is administration of items to a sample of test takers from the target population. The best way to do this kind of field-testing is to embed field-test items in the actual test. Field-testing new items alone is not a good idea, because test takers will know that the test is not a real test. The information that results is likely to be invalid. The second aspect is to interview test takers as they self-administer the item. This method has been referred to as *think-aloud* and *cognitive lab*.

Validity Evidence for Item Validation

As noted in this chapter and again throughout this book, the performance of certain activities is an important source of validity evidence. Instead of statistical results, we are looking at procedural results. Not only is it important to perform the many steps described in this chapter, but documenting these activities is also important. Testing programs highly concerned about validity should archive all reports involving any of these steps and make it known in their technical reports that such activities have been completed. Also using an appropriate website is another effective way to show that these important item development activities have been done.

Summary

This chapter gives a brief overview of the many important steps involved in item development. Many of these steps by virtue of their completion present validity evidence (Downing &

Haladyna, 1997). This is procedural evidence: events recorded that show that vital actions were taken to validate items. Because these events comprise an important source of item validity evidence, documenting when these events were held and completed is important. Other chapters provide great detail about some of these procedures as noted in this chapter.

3
Content and Cognitive Demand of Test Items

Overview

Content-related validity evidence is a major consideration in any validation of a test score interpretation or use (Kane, 2006a). Content also plays a very important role in item development and validation. A panel of highly qualified subject-matter experts (SMEs) is critical in establishing a body of content-related validity evidence. Their expertise and judgment comprise the basis for the validity argument supporting the content of a test. Thus, the focus of this chapter is on developing content-related item validity evidence as part of item validation. Two major issues in this quest for content-related validity evidence and item validation are content and cognitive demand.

Content refers to knowledge, skills, and abilities, which were briefly defined in chapter 1. The use of knowledge and skills in performing a complex behavior is characteristic of a cognitive ability. In chapter 1, we used writing in the language arts curriculum, and dentistry, a professional competence, as examples to illustrate two very different cognitive abilities.

Cognitive demand refers to the *expected* mental complexity involved when a test item is administered to a *typical* test taker. Recalling knowledge is the simplest form of cognitive demand. Comprehending or understanding knowledge is a slightly higher cognitive demand. Tasks that involve the complex use of knowledge and skills are the highest type of cognitive demand. Examples include solving a problem, writing poetry, or completing a science project. Any cognitive demand depends on the nature of the task and the instructional history of the test taker. Thus, no item or objective has an absolute cognitive demand. The cognitive demand that we assign to a test item is only a best guess based on the speculation about the cognitive process needed to respond to the test item correctly. This speculation considers the typical test taker, not necessarily the very advanced learner or someone who has not yet learned that content.

The first part of this chapter discusses how cognitive psychology and measurement theorists led an effort to improve the measurement of cognitive abilities, such as that found in schools and professions. Then we discuss limitations of the most popular cognitive taxonomy for classifying types of cognitive demand. In the next section, we present a simplified cognitive taxonomy that draws from our understanding of knowledge, skills, and abilities. When we have organized the target domain and the universe of generalization for content and cognitive demand, a set of item and test specifications are created. Other terms used for the item and test specifications document are *two-way grid* or *test blueprint* (e.g. Gronlund & Waugh, 2009; Linn & Miller, 2005; Thorndike & Thorndike-Christ, 2010). This chapter's final section identifies the content-related validity evidence needed in item validation. Naturally, content and cognitive demand are the mainstays of this evidence.

The Role of Cognitive Psychology in the Measurement of Achievement

Behavioral learning theory reached its zenith in the mid-20th century as an influence on teaching, learning and measuring achievement. Criterion-referenced tests, mastery learning, behavioral/instructional objectives, Bloom's cognitive taxonomy, and learning domains are the byproducts of behavioral learning theory. Vestiges of behavioral learning theory remain in state content standards and k-12 testing programs as instructional objectives. Teachers are instructed and expected to teach to these standards rather than develop cognitive abilities. However, these remnants of behaviorism in education and professional development are disappearing as cognitive psychology takes on a greater role in construct definition, item development and validation, test design and development, reporting, and validation.

The idea of construct theory appears in more previous and contemporary discussions of validity (Cronbach & Meehl, 1955; Messick, 1989). Cognitive psychology's contribution to improving the validity of measurement of achievement continues to be the object of considerable study and research (Ferrera, 2006a, 2006b; Gierl & Cui, 2008; Gorin, 2006; Lane & Stone, 2006; Leighton & Gierl, 2007; Mislevy, 2006; Snow & Lohman, 1989; Sugrue, 1995). Without doubt, cognitive learning theory is dominant and drives current thinking about learning and measuring achievement.

Cognitive learning theory is hardly a refined, cohesive, unified science. Despite the lack of a single, unified theory of student learning, a consensus exists about the importance of defining and developing cognitive abilities (Lohman, 1993; Messick, 1984; Mislevy, 2006; Snow & Lohman, 1989; Sternberg, 1998). As noted in chapter 1, a cognitive ability is an intricate mental structure that consists of a domain of tasks. Each task requires the use of knowledge and skills in complex ways for a desired purpose. Each cognitive ability is very slow-growing. The teaching and testing of a cognitive ability involve more than simply acquiring knowledge and skills (Frederiksen, 1984; Lane, 2004; Mislevy, 2008; Nichols & Sugrue, 1999; Sugrue, 1995). *Test Theory for a New Generation of Tests* provided one of the best accounts of emerging thinking about cognitive processes and test item design (Frederiksen, Mislevy & Bejar, 1993).

Some major concepts and distinctions from cognitive psychology introduced in chapter 1 figure very importantly in this chapter:

1. Developing a learner's cognitive abilities is the goal of most k-12 instruction and professional education. We want students who have adequately developing reading, writing, speaking, listening and mathematical and scientific problem-solving abilities. Other abilities are developed in schools, including critical/analytical thinking and various creative abilities. In the training of any profession, candidates for licensure or certification must acquire knowledge and skills and put these to work on more complex tasks that they encounter in their profession.
2. We have two different kinds of domains of learning. The first is more traditional. It consists of knowledge and skills organized into a hierarchy (Messick, 1994; Sugrue, 1995). This domain is based on behavioral learning theory. The terms *criterion-referenced* and *domain-referenced* have been often used to describe tests designed to sample from this domain. The second type of domain is a collection of performed tasks that require the complex use of knowledge and skills. Messick referred to this type as task-based and the former as construct-based. Cognitive learning theory identifies closely with this second type of domain. In chapter 1 and this chapter, we have used two kinds of cognitive abilities, writing and dental competence, to illustrate features of this kind of domain.
3. Regarding this second type of learning domain, Kane (2006a, 2006b) stated that a *target domain* is the reification of the domain of tasks to be performed. For instance, a

target domain for writing includes all possible writing tasks that we might encounter in a lifetime. A target score is the score a test taker might achieve if all items/tasks in this domain were administered. The target domain is a heuristic device, and the target score is hypothetical.

4. Realistically, the *universe of generalization* represents those test tasks/items that might be included on any test of that construct. The universe score is the true score—the score obtained if all items in the universe of generalization were administered. Ideally and hypothetically, the universe score is perfectly correlated with the target score. The judged correspondence between the tasks in the target domain and the tasks in the universe of generalization is an important outcome in the development of content-related validity evidence.
5. A validly interpreted test score should result in designing a test that is a representative sample from the universe of generalization. The item and test specifications document is the device we use for this end.
6. Without doubt, sponsors of virtually all types of tests have wanted to engage test takers in test items with higher cognitive demand. Research and surveys continually show that tests have far too many items calling for low cognitive demand. Thus, test developers are concerned about including objectives that have a higher cognitive demand and items that match their content standards or competencies.
7. As stated previously, the cognitive demand of any item is a function of the developmental level of the test taker and the design of the item. An expert learner works mainly from memory by recalling an experience that fits a task/test item demand. The cognitive demand is relatively simple. Perhaps it is recall or recognition. A novice learner may have to invoke more complex strategies to respond to the task/test item because of a lack of knowledge and skills and experience with the task. Thus, the test item will have a high cognitive demand for a novice learner. We can only speculate about cognitive demand for any test taker, unless we use a think-aloud procedure to see what the test taker was thinking when responding to the item.
8. Identifying the content of each item is crucial. Knowing its cognitive demand for test takers of an average or desired developmental level is also important. Not only is instruction improved by knowing about content and cognitive demand, but testing is focused exactly on what the construct represents.

The main issue in this chapter is to show that the universe of generalization has a high degree of fidelity with the target domain. Then another goal is to ensure that any test is an adequate, representative sample of test items from this universe of generalization. The item and test specifications document is the current technology for designing this kind of test. However, cognitive psychologists have been working on alternative methods for test design that avoid test specifications and focus on the cognitive ability itself (see Mislevy, 2006, for an example of a cognitively based approach to test design).

Both Messick (1989) and Kane (2006b) have argued that content-related validity evidence should address important concepts. These include content relevance, content representativeness, dimensionality/structure, adequacy of the item pool, and internal and external aspects of item response and test score structure. Later in this chapter, we will address the assembling of evidence to support item validation as for content and cognitive demand.

The next section deals with the vexing problem of ascertaining the cognitive demand represented in the test item, given the repeated caveats about the variability that naturally exists among test takers with cognitive demand for any item.

Limitations of the Cognitive Taxonomy for Classifying Cognitive Demand

The cognitive taxonomy was conceived to help educators classify instructional objectives, but it is also used to classify test items. The taxonomy is the result of a project headed by the highly respected educator Benjamin Bloom and his colleagues (Bloom, Engelhart, Furst, Hill, & Kratwohl, 1956). Since its publication, the taxonomy has enjoyed enormous popularity. A Google visit will yield nearly a million hits on the taxonomy. There is little doubt that the taxonomy has inspired educators to create instructional objectives, teach accordingly, and measure the learning that has a high cognitive demand. Without reservation, the taxonomy has affected educators.

The Current Cognitive Taxonomy

The current cognitive taxonomy consists of six distinct categories: knowledge, comprehension, application, analysis, *evaluation, and create* (Anderson, Krathwohl, Airasian, Cruikshank, Mayer, Pintrich, Raths, & Wittrock, 2001; Anderson & Sosniak, 1994). The original taxonomy was slightly modified: the new taxonomy reversed the order for *synthesis* and *evaluation* and changed *synthesis* to *create*. The first two categories of the taxonomy, knowledge and comprehension, are the most recognizable and useful. The other four categories represent more complex forms of thinking where knowledge and skills are applied in a situation to achieve an intended outcome. In his interesting book *Classroom Questions*, Sanders (1966) provided many examples of test items based on this cognitive taxonomy.

In the item and test specifications document, a cognitive demand dimension might be added to a content dimension to provide a two-dimensional array of content and cognitive demand for test design. Most classroom assessment textbooks provide an explanation of the cognitive taxonomy and recommend that classroom tests be organized using this item and test specifications document.

Research on the Validity of the Cognitive Taxonomy

Since the introduction of the cognitive taxonomy, research on the validity of the taxonomy has been inadequate. A review of research current to 1978 resulted in this conclusion:

> As a final assessment of the validity of the claims concerning the psychological properties of the taxonomy, it is perhaps fairest to say that the picture is uncertain. No one has been able to demonstrate that these properties do not exist. (Seddon, 1978, p. 321)

A review by Kreitzer and Madaus (1994) drew a similar conclusion.

Any research on the cognitive taxonomy can be classified in one of four categories. Studies of internal structure of content should show that the six categories of the cognitive taxonomy when applied to test item responses are real and identifiable. Studies of classification consistency of instructional objectives or test items should show how well the taxonomy is understood and works with SMEs. If the taxonomy is useful, we would like to see high degrees of classification consistency. A newer type of study involves interviews of students who report on the cognitive demand of a test item. The fourth type of study is critical analysis. This kind of study is done by philosophers and psychologists concerned about learning. As most philosophers and psychologists have a constructivist or cognitive science orientation, their essays and other writing reflect a contemporary view of higher level thinking that is critical of the cognitive taxonomy, which is rooted in behavioral learning theory.

Studies of Internal Structure

A study reported by Stoker and Kropp (1964) reported that knowledge/comprehension levels of the taxonomy formed a common factor, and they concluded that there was support for the

hierarchical structure of the data. A study by Miller, Snowman, and O'Hara (1979) using the same data and different methods of analysis concluded that the structure of the data resembled fluid and crystallized intelligence. Another reanalysis of the Stoker/Kropp data by Hill and McGaw (1981) using another method of analysis concluded some support for higher-order thinking. Kunen, Cohen, and Solman (1981) found some evidence for a hierarchy but thought that the evaluation category was misplaced. We have no more current studies to report of the structure of data generated for the cognitive taxonomy.

The study of internal structure in test item response data is hopelessly muddled by the fact that the cognitive demand of any item is a joint function of the item and the developmental level of the student being tested. As noted previously in this chapter, a novice level test taker is likely to be a low performer and will encounter the task in the test item with a more complex form of cognitive behavior. An expert test taker who performs at a high level will simply recall from memory. Thus, it is no wonder that these studies fail to produce item response patterns resembling the cognitive taxonomy. If the learning history of each test taker were known, such studies might reveal more about the veracity of the cognitive taxonomy.

Studies of Classification Consistency

A first study of the classification consistency of test items using the cognitive taxonomy was done by Stoker and Kropp (1964). Although they reported very little data, they observed a high degree of classification consistency. Researchers working with the faculty and students in college-level biology reported success with the use of the taxonomy (Crowe, Dirks, & Wenderoth, 2008). They also reported success with teaching students to classify and write test items using the taxonomy. Another study of classification consistency was conducted in two veterinary courses using SMEs and non SMEs (van Hoeij, Haarhuis, Wierstra, & van Beukelen, 2004). Degrees of agreement ranged from 34% to 77%. Kappa coefficients of agreement that were weighted ranged from .12 to .60. Despite the low degree of agreement, the researchers reported that the use of the taxonomy had a positive influence on the teaching staff regarding the importance of developing test items with higher cognitive demand. Two studies in medical education addressed the issue of the relationship of different levels of student learning using the cognitive taxonomy as predictors of clinical performance (Norcini, Swanson, Grosso, Shea, & Webster, 1984; Peitzman, Nieman, & Gracely, 1990). In both studies no discernible difference was found between lower- and higher-level learning in relationship to clinical performance. In another medical school study, researchers categorized test items by the basic distinctions in the cognitive taxonomy: lower, moderate complexity, and higher order (Cunnington, Norman, Blake, Dauphinee, & Blackmore, 1996). Agreement among SMEs on this three-category classification exercise was low. The generalizability coefficient was .69. There was no performance difference. They concluded: "Thus the taxonomic level of an item is not absolute, but is related to the knowledge and understanding of the examinee" (Cunningham et al., 1996, p. S32). If the taxonomy is to be a useful tool in classroom and large-scale testing of a curriculum or a set of professional competencies, SMEs need to classify both objectives and test items with a high degree of consistency. There is scant research on classification consistency using the cognitive taxonomy, and what little is reported is not encouraging.

Studies Involving Interviews With Test Takers

Another way to uncover the cognitive demand of instructional objectives or test items is to interview students. Hibbison (1991) performed an experiment with his college-level students. After they took a 40-item selected-response test, he asked five students what they were thinking. He found 27 different inferences, which he attributed to meta-cognitive, cognitive, and affective

origins. In a study by Gierl (1997) involving seventh-graders' mathematics achievement test items, the correspondence between the expected classifications as judged by item writers and students was 54%. He concluded that the taxonomy does not help item writers to anticipate the cognitive demand of students. A very interesting study by Solano-Flores and Li (2009) used cognitive interviews with different cultural groups in the fifth grade for mathematics. They found that students had different perspectives for each item that influenced the way they approached each item.

> What studies are beginning to show is that students can generate a correct answer using patterns of thought that are unrelated to knowledge and skills targeted by the test item. Genuine domain mastery or competence can be usurped by testwise strategies and alternative knowledge not specifically targeted by the test item. (Leighton & Gierl, 1997, p. 5)

Thus, when a team of SMEs makes a claim for an item for its cognitive demand, the actual thought process employed by each test taker varies considerably with learners. Every test item has a personal context with the student, and that fact influences how they perform. Sugrue (1995) also provided greater understanding of the problem of determining the cognitive demand in a student's response to a test item. As students vary in their instructional history, mathematics problem-solving cannot be treated as a unified construct. Her analysis led her to conclude that students need a more personalized approach to something as complex as problem-solving.

Critical Analyses

Critical analyses of the cognitive taxonomy were of great interest long ago (Furst, 1981; Poole, 1971, 1972). Furst's findings generally supported the use of the taxonomy, although he acknowledged and cited instances where users had simplified the taxonomy to three categories for convenience of use. Cognitive learning and constructivist theorists consider the cognitive taxonomy to be outdated (e.g. Bereiter & Scardamalia, 1998). These authors have adopted the position reported in studies of student think-aloud; cognitive demand depends on the developmental learning level of the student. No item has a natural cognitive demand. Other philosophers and cognitive psychologists have been critical of the cognitive taxonomy and presented arguments in favor of the interpretation of cognitive demand based on the interaction of each student and the item (Ennis, 1989, 1993; Lewis & Smith, 1993; Mayer, 2002). Much of higher-level thinking consists of structured and ill-structured problems that require the application of knowledge and skills. A cognitive task analysis performed by SMEs or as reported by students performing the task is the best way to uncover a cognitive demand. Clearly, the direction taken by contemporary learning theorists is away from the mental-filing-cabinet approach of the cognitive taxonomy. Because we understand more about complex thinking, behaviorism does not explain it very well, and cognitive psychology seems better suited for modeling complex thinking.

Conclusion About the Cognitive Taxonomy

Despite the cognitive taxonomy's considerable reputation, it has not been adequately validated. Since the review by Seddon (1978), subsequent research has not established the cognitive taxonomy as a useful tool in item development and validation. Studies of internal structure that reflects the six categories of the cognitive taxonomy have failed to yield evidence. SMEs have typically failed to master the classification of items and instructional objectives into the six categories. Think-aloud/cognitive interview studies reveal the hopelessness of pinpointing any item as having a specific cognitive demand due to the variety of developmental levels in any group of test takers. Critical analyses by contemporary learning theorists reject the taxonomy. Lewis and Smith (1993) quote one critic as calling the task of classifying cognitive behavior a

"conceptual swamp." To illustrate, we have a plethora of terms taken from various sources that convey higher-level thinking:

critical thinking	thinking skills
complex thinking	inferences
information processing	higher mental processes
habits of mind	metacognition
scientific problem-solving	strategic thinking
mathematical problem-solving	inference
higher-order thinking skills	analytical thinking
evaluation	creative thinking
comparative investigation	declarative knowledge
application	procedural knowledge
synthesis	reasoning
creation	reproductive thinking
analysis	higher-level thinking

We have a very loose use of terminology without adequate definition. It is no wonder that no taxonomy will ever work until any category of higher-level thinking is more adequately defined and terminology is standardized. This collection of ill-defined terms is not the fault of behavior learning theorists but more a problem for all learning theorists and the testing specialists who carry out various taxonomic schemes.

In its current form, the cognitive taxonomy is far too complex to be workable in the organization of curriculum, professional competence, or a cognitive ability such as writing.

The idea of simplifying the cognitive taxonomy has merit because there is more consensus about the first two levels of the cognitive taxonomy, and the application of knowledge and skills in some complex ways represents a third type of cognitive demand that is a catch-all category for the complex use of knowledge and skills.

Recommendation for Classifying Content and Cognitive Demand

This section presents a simplified cognitive taxonomy. The recommendations are compatible with the idea of the cognitive ability as the focus of all instruction. Suggestions for simple taxonomies of the type proposed here have existed (Ayers, 2001; Collins, 2006; Furst, 1981). This recommendation is consistent with the call for simplicity.

Knowledge, Skills, Abilities

The phrase *knowledge, skills, and abilities (KSAs)* is often used to capture the range of content and cognitive behaviors found in learning and testing. In this section, the three concepts have very specific meanings regarding content and cognitive demand. The method for organizing content and cognitive demand is straightforward. We adhere to the goal of developing cognitive abilities. Knowledge and skills are important aspects of cognition, but both play supportive roles in the performance of complex tasks that have a high cognitive demand.

Knowledge

Borrowing from the definition of knowledge proposed by David Merrill (1994), all knowledge can be classified as either a fact, concept, principle, or procedure. All knowledge is subject to

three types of cognitive demand: (a) recall/recognition, (b) comprehension/understanding, and (c) application in a complex task.

Fact A fact is a truth known by experience or observation. A fact is a statement that is indisputable. In this chapter and test content, facts are established by SMEs. Most facts are committed to memory and can be recalled or recognized in a test. Facts should be distinguished from opinions. Opinions have different points of view and rationales for each opinion. Most elementary school curricula help learners determine the difference between a fact and an opinion. Regarding the cognitive demand, facts may be recalled and facts may be understood. Facts may also be used in a more complex task as part of an argument, in the development of a solution to a problem, or in some creative way.

For writing ability some facts are:

1. A period is placed as the end of declarative sentence.
2. A compound sentence is punctuated with a comma after the first part.
3. The word "fact" is spelled f-a-c-t.
4. A proper noun begins with a capital letter.
5. The alphabet.

Concept A concept is an idea of something formed by mentally combining all its characteristics or particulars. A concept has a name, distinguishing characteristics, and examples and non-examples. As with any fact, we can recall the definition of a concept, understand or comprehend the meaning of a concept, or apply a concept for a more complex task. A concept can be defined literally by recalling or recognizing a written definition. Or a concept can be understood by presenting it in a paraphrased version and asking the learner to identify or create a response that exhibits understanding. Another way to test for understanding/comprehension is to present examples and non-examples that have not been previously introduced. In the performance of a complex task, the learner may use the concept with other concepts to solve a problem, think critically, or analyze.

For writing ability, some concepts are:

1. Persuasive writing
2. Punctuation
3. Spelling
4. Grammar
5. Word

Principle A principle is an accepted rule of action or conduct, a fundamental, primary, or general law or truth from which others are derived, a fundamental doctrine or tenet. Usually, we recognize a principle as a statement of relation among two or more concepts. The principle can be absolute/axiomatic: hot air rises. Or the principle can be probabilistic: on 1,000 flips of the coin, the likely result for heads is about 50%.

Some examples of principles are:

1. The chance of fatal injury when a passenger has a fastened seatbelt is less than if the passenger had an unfastened seatbelt.
2. The origin of humans is a complex evolutionary story.
3. Smoking causes respiratory and other illnesses.
4. A paragraph should begin with a topic sentence.
5. When water evaporates, it carries heat with it.

Procedure A procedure is an observable physical or mental course of action that has an intended result. Although procedures are associated with skills, a fundamental requirement of performing any skill is knowledge of the procedure. Examples of procedures include knowing how to:

1. Unlock a door.
2. Wash a car.
3. Estimate the amount of paint to buy to paint a bedroom.
4. Water your vegetable garden.
5. Turn on your computer.

Use of Knowledge in Developing an Ability and In Test Design This four-category organization of knowledge helps item writers better understand the variety of content of knowledge that is available when creating a test item. An item need not be classified as either fact, concept, principle, or procedure. However, as a construct is being defined, the atomistic analysis of knowledge should lead to more precise teaching and more effective learning that is guided by more valid testing for formative and summative purposes.

Three Cognitive Demands of Knowledge

With these four types of knowledge, we can ask a learner to:

1. recall or recognize knowledge verbatim.
2. comprehend or understand knowledge.
3. use knowledge in a complex task (that reflects an ability).

Table 3.1 shows the three types of cognitive demand for different content.

Table 3.1 Three Types of Cognitive Demand for Knowledge

	Cognitive Task		
Topic	**Recall of Knowledge**	**Comprehend/Understand Knowledge**	**Use Knowledge**
Periods	A period ends a declarative sentence.	How do we use a period in writing? Why?	In writing a persuasive essay, the writer punctuates the ending of each sentence correctly.
Triangles	The sum of angles in a triangle is 180 degrees.	Given a variety of geometric figures found in nature and society, identify a triangle.	Develop a proof that shows the sum of angles for any triangle is 180 degrees.
Gas	A gas takes the shape of its container.	Why does a gas take the shape of its container?	You determine that a toxic gas has been filling your kitchen. What is your next course of action?
Compass	A compass points to the north.	Why does a compass point to the north?	You are lost in the woods, and you find a compass in your backpack.

These examples show that for different kinds of content we essentially have three cognitive demands: recall of knowledge, understanding of knowledge and the use of knowledge for a more complex task. Note that this organization of cognitive demand is very much like the traditional cognitive taxonomy, but the application of knowledge applies to the four types of higher-level thinking found in the cognitive taxonomy.

Skill

A skill is a performed act. The structure of any skill is simple. Some skills consist of a singular act whereas other skills are procedures involving two or more steps. The distinction between a skill

and the performance of a complex task representing an ability is arbitrary, but the latter distinction reflects one task in the universe of generalization for a cognitive ability. The performance of these complex tasks involves more than just performing a skill. A committee of SMEs is best suited to judge whether a performance is an instance of a skill or an instance of a task from the universe of generalization for an ability.

Any skill has three types of cognitive demand:

1. We can have knowledge of a skill. We can memorize a procedure or understand a procedure. We can know how to spell a word or how we should punctuate sentences. Knowing is good, but doing is different.
2. We can perform the skill. The performance requires knowledge, so we can agree that a hierarchical structure exists. One has to know how to do something before one does it well. An editor uses editing skills often, such as correcting spelling or punctuation.
3. The skill can be performed as a component part of a procedure associated with a task representing a cognitive ability. For instance, in the writing of a book, the performance of many editing skills is needed.

Table 3.2 shows the progression of knowledge of a skill to the performance of the skill in an isolated way to the use of a skill in a more complex task.

Table 3.2 Three Types of Cognitive Demand for a Skill

	Cognitive Task		
Topic	**Knowledge of a Skill**	**Performance of a Skill**	**Use of a Skill in a More Complex Task**
Punctuation	You should end a sentence with a period. You should end a question with a question mark. You should end an exclamatory sentence with an exclamation mark. A phrase does not have ending punctuation.	A list of sentences is provided and you need to provide ending punctuation.	In an essay you have written, you have correctly punctuated the end of each sentence.
Absolute value	Express that the distance between two numbers is the absolute value of their difference. Whether the difference is positive or negative, the absolute difference is unsigned value.	Given a series of absolute statements, calculate the absolute value. For example: $\|9–12\|$	You are driving from one city to another city on Interstate 999. Rose City is at mile marker 232. Geranium City is at mile marker 311. How far is Geranium City from Rose City?
Metaphor	Identify a metaphor in literature.	Below are ten statements: Mark A if an example of a metaphor and Mark B if not an example of a metaphor.	Discuss the use of metaphors in a short story you have written.

Cognitive Ability

As presented in chapter 1, a cognitive ability is a mental capacity to perform any task from a domain of tasks. Each cognitive ability is represented by a target domain that contains a population of complex tasks. In mathematics education, this target domain might be all problems requiring mathematics that we encounter in our daily lives. Any complex task in mathematics will require that knowledge and skill be applied in a complex way. Evidence of its complexity comes from a cognitive task analysis conducted by SMEs or via discussions with targeted learners.